Semantic Segmentation of Aerial Imagery Deep Machine Learning

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Introduction, Data and Model Architecture

A deep learning project utilizing the U-Net architecture [4] for semantic segmentation of aerial imagery over the city of Potsdam, Germany.

A functional semantic segmentation model applied to aerial images will make it possible to classify objects and surfaces such as trees, roads, cars and buildings. This can be useful for tasks such as urban mapping, environmental monitoring and land cover analysis [2].

The data utilized are very high resolution (VHR) True Ortho Photos over Potsdam and comes from ISPRS Potsdam dataset [1].

Segmentation of VHR images is particularly challenging as small objects and small-scale surface texture become visible, leading to high within-class variability and at the same time low inter-class differences [3].

The data:

- RGB image tiles and RGB label tiles, `(6000 x 6000 x 3)`
- Imbalanced data of six different classes

Impervious surfaces	(white)	30.26% of training data
Building	(dark blue)	27.32% of training data
Low vegetation	(cyan)	23.21% of training data
Tree	(green)	12.96% of training data
Car	(yellow)	1.77% of training data
Clutter/background	(red)	4.48% of training data

- Training data, 16 image tiles divided into 1883 patches of `(512 x 512 x 3)` pixels
- Validation data, 5 tiles → 605 patches
- Test data, 9 tiles → 1089 patches
- Label patches transformed into integer label patches

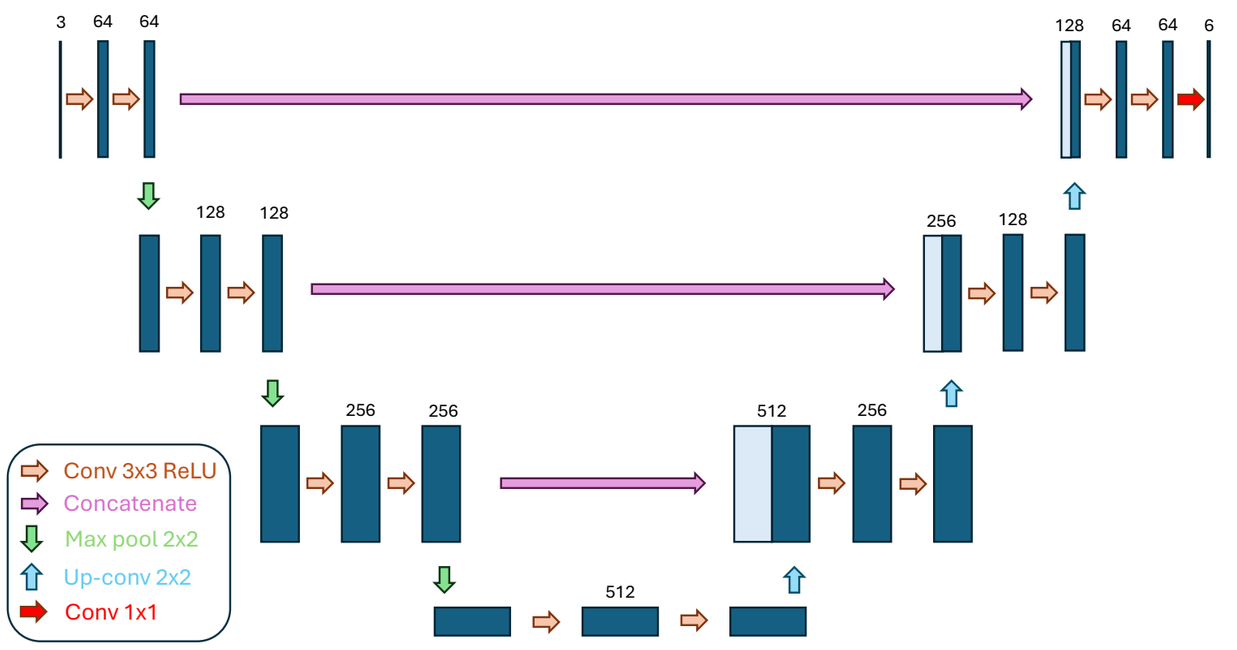


Figure 1: Our U-Net model architecture

The U-Net model architecture:

- The U-Net architecture was introduced as a way to better make use of limited availability of labeled data for biomedical image segmentation and showed great performance [4]
- An influential encoder-decoder segmentation architecture
- It consists of a contracting path and an expansive path
  - Contracting path with convolutions to increase depth and capture context, and maxpooling to reduce `(H, W)` dimensions
  - Expansive path with upsampling to increase `(H, W)` dimensions, linked with skip-connections to get global context from earlier feature maps
  - A final layer with a `(1 x 1)` convolution with six channels, to produce logits for each class and each pixel
- Avoids the use of fully connected layers, reducing the number of trainable parameters for faster training
- Aerial images of cities have no inherent direction, similarly to many biomedical image segmentation problems

Learning approach

- Scale image data to range $[-1, 1]$
- Data augmentation
  - Horizontal flips, vertical flips, 90 degree rotations
- Adaptive learning rate
  - ReduceLROnPlateau
- Loss function
  - Sum of Dice loss and Focal loss
- Save model parameters when
  - validation loss is lower than previous best and
  - validation mean IoU is higher than previous best

Performance on Test data

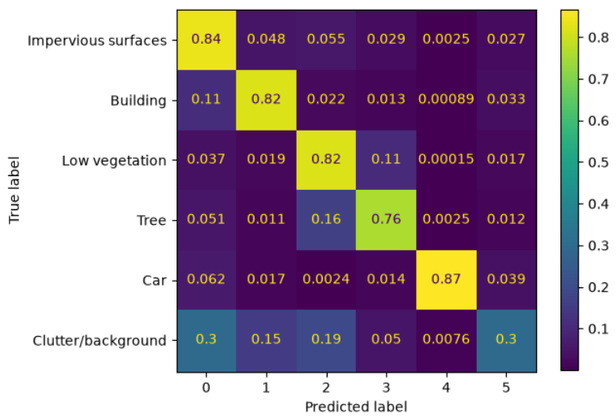


Figure 2: Confusion matrix, performance on test data

- mean IoU: 0.615
- Per-class IoU
  - Impervious surfaces: 0.7135
  - Building: 0.7351
  - Low vegetation: 0.6341
  - Tree: 0.6093
  - Car: 0.7929
  - Clutter/background: 0.2021
- Macro F1: 0.739
- Accuracy: 0.931

Example predictions

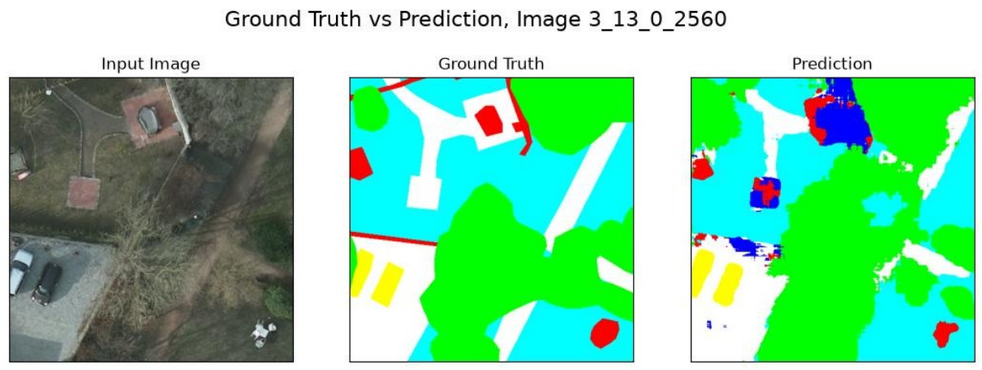


Figure 3: Prediction

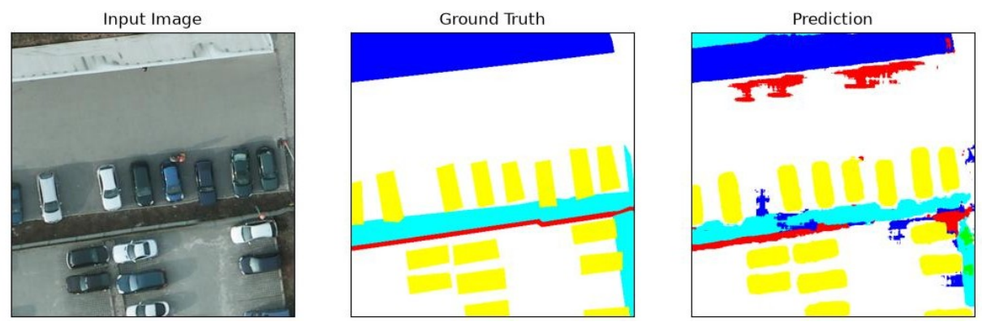


Figure 4: Prediction

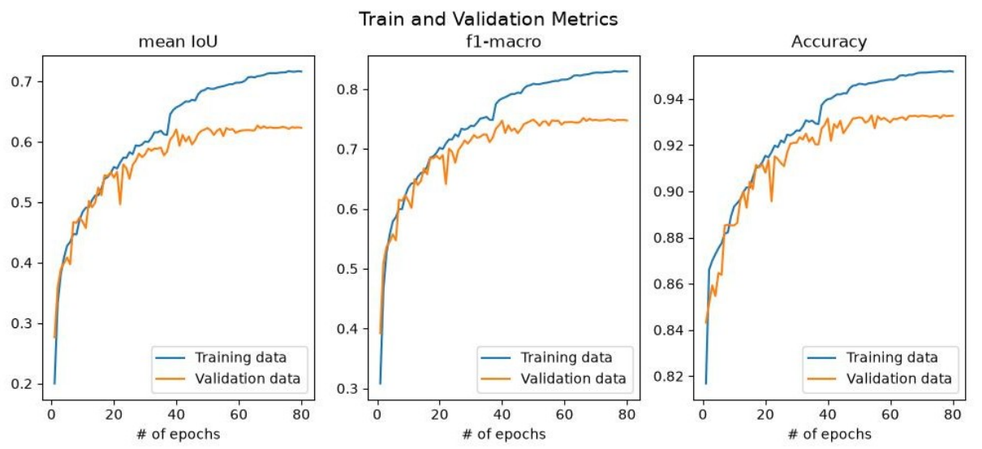


Figure 5: Training and Validation metrics

Conclusions:

- Validation metrics plateaus after about 50 epochs
- Best model parameters saved after epoch 66
- Cars are the smallest class by number of pixels. Per-class IoU tells our model learns to segment cars well
- Worst performance for the clutter/background class. Not surprising since this class constitutes of many different real world objects

References

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[3] D. Marmanis, J. D. Wegner, S. Galliani, K. Schindler, M. Datcu, and U. Stilla.
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